



Grouping Methods

Applied Machine Learning

Clustering Analysis

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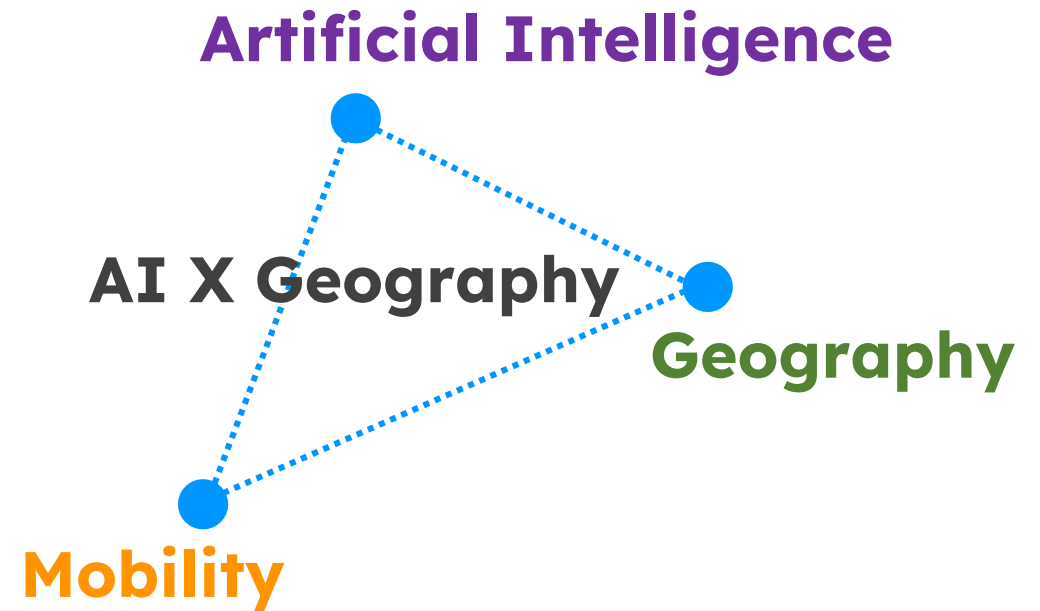
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Outline

#GeoAI #Mobility #Geography #STDS

- 01 Clustering Analysis
- 02 Evaluation Metrics
- 03 K-means Clustering
- 04 Hierarchical Clustering
- 05 DBScan
- 06 HDBScan
- 07 Optics
- 08 Summary



Clustering Analysis

From predicting known classes to discovering unknown structures, we want to know which observations are much more similar each other.

In classification, we already know **what we are looking for**.

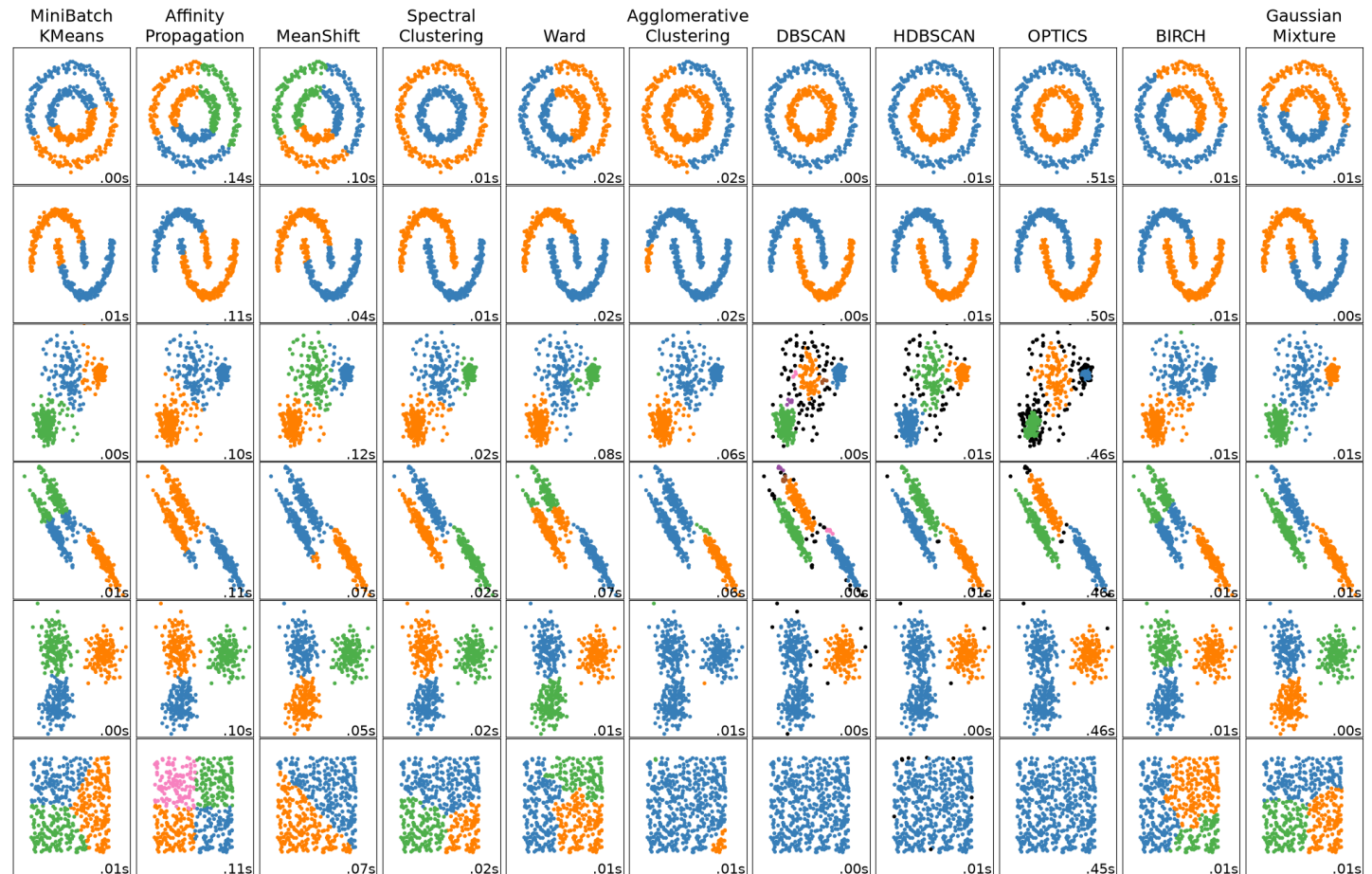
In clustering, we ask the data **what structures are hidden within it**.

Application:

1. **Discover hidden structures:** Identify natural groups that are not predefined.
2. **Understand heterogeneous populations:** Reveal different types of observations within seemingly similar data.
3. **Simplify complex datasets:** Summarize thousands or millions of observations into meaningful groups.
4. **Support downstream decisions:** Use discovered groups for segmentation, anomaly detection, recommendation, spatial analysis, or further modeling.

Clustering Analysis

Types of Clustering Analysis in sklearn

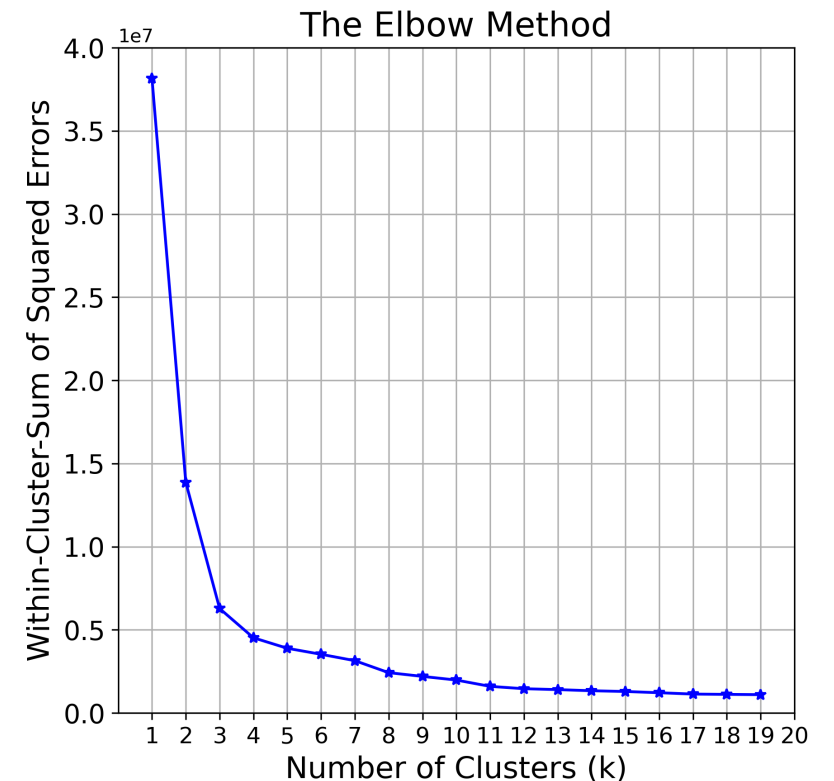


Evaluation Metrics

Within-Cluster Sum of Squares (WCSS), measures the sum of squared distances between each data point and the centroid of its assigned cluster. The centroid is the mean position of all the points in the cluster.

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

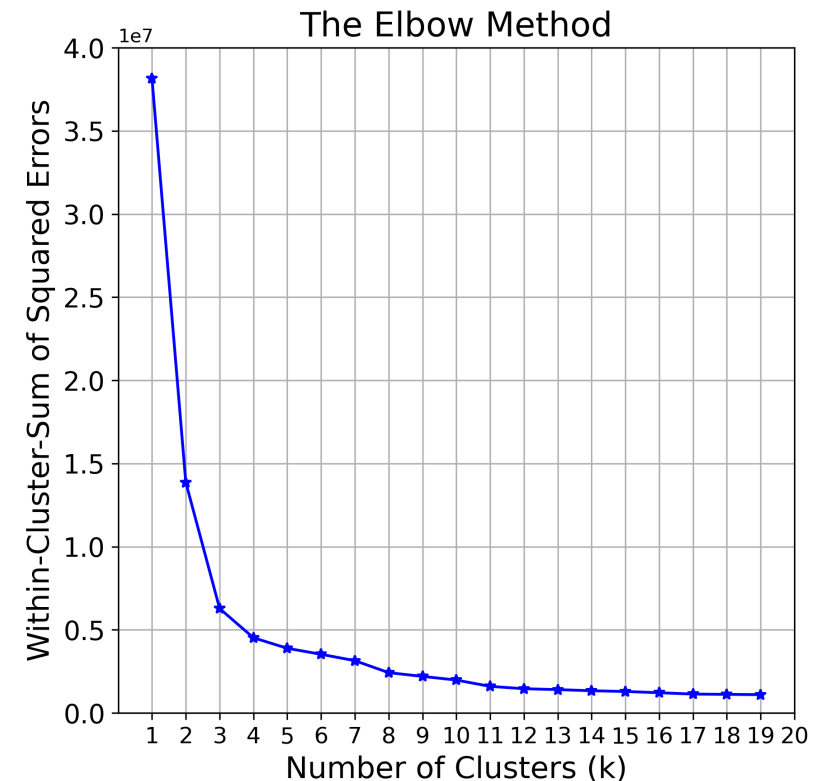
k is the number of cluster; C_i represents the i th cluster; x denotes the data points in cluster C_i ; μ_i is the centroid of cluster C_i ; $\|x - \mu_i\|^2$ is the squared Euclidean distance between point x and centroid μ_i .



Evaluation Metrics

Calculate the WCSS for different values of k , and choose the k for which WSS first starts to diminish. In the plot of WCSS-versus- k , this is visible as an elbow.

The Squared Error for each point is the square of the distance of the point from its representation, i.e., its predicted cluster center. Any distance metric like Euclidean or Manhattan Distance can be used.

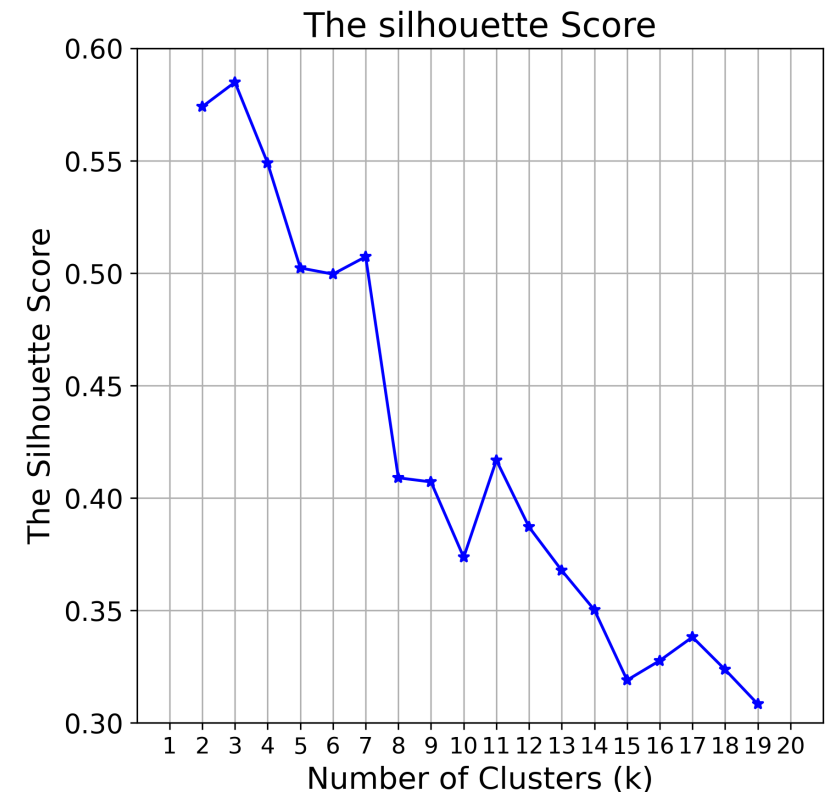


Evaluation Metrics

The **silhouette value** measures how similar a point is to its own cluster (cohesion) compared to other clusters (separation).

The range of the Silhouette value is between $+1$ and -1 . A high value is desirable and indicates that the point is placed in the correct cluster.

If many points have a negative Silhouette value, it may indicate that we have created too many or too few clusters.

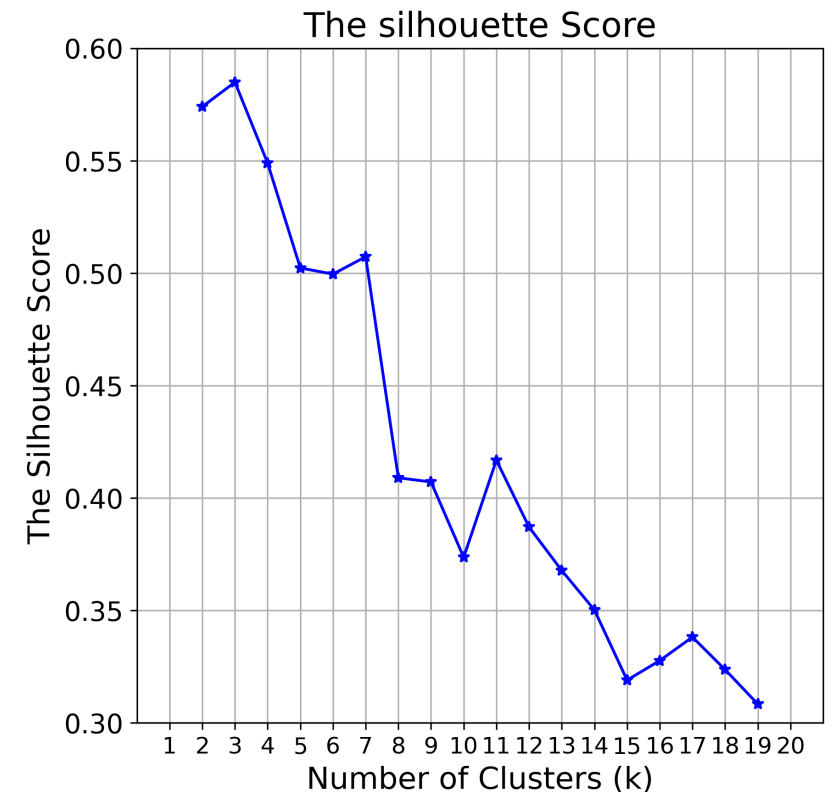


Evaluation Metrics

The Silhouette value $s(i)$ for each data point i is defined as follows:

$$\begin{cases} s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \text{ if } |C_i| > 1 \\ s(i) = 0, \text{ if } |C_i| = 1 \end{cases}$$

$s(i)$ is defined to be zero if i is the only point in the cluster. This is to prevent the number of clusters from increasing significantly with many single-point clusters.



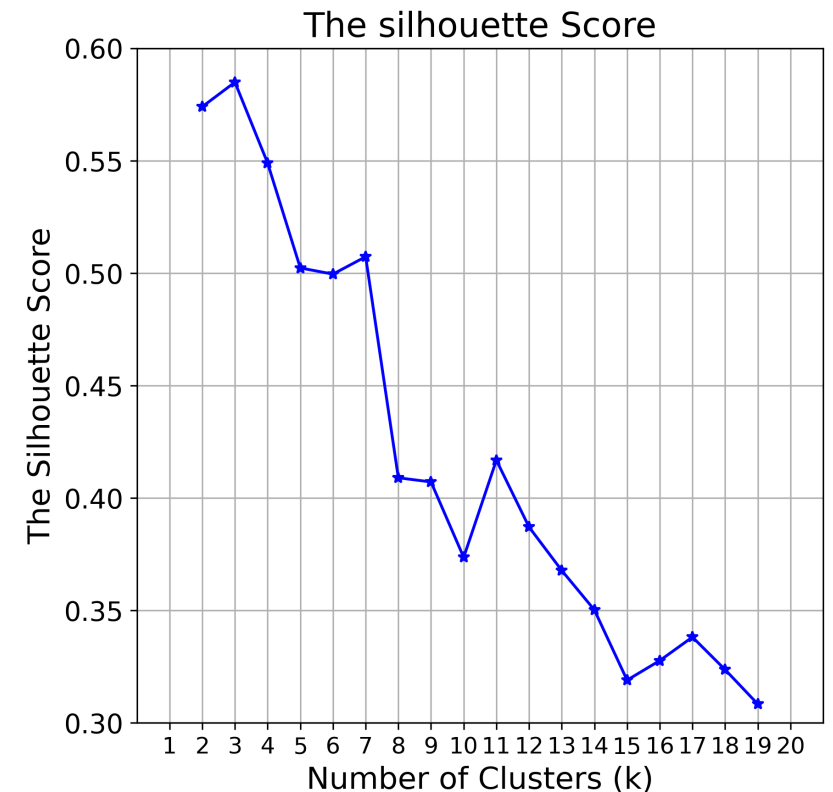
Evaluation Metrics

Here, $a(i)$ is the measure of similarity of the point i to its own cluster. It is measured as the average distance of i from other points in the cluster.

$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, i \neq j} d(i, j)$$

Similarly, $b(i)$ is the measure of dissimilarity of i from points in their clusters.

$$b(i) = \min_{i \neq j} \frac{1}{|C_j|} \sum_{j \neq C_j} d(i, j)$$



K-means Clustering



k-means clustering is a method of vector quantization, originally from signal processing, that aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest mean (cluster centers or cluster centroid).

The unsupervised k-means algorithm has a loose relationship to the k-nearest neighbor classifier, a popular supervised machine learning technique for classification that is often confused with k-means due to the name.

K-means Clustering

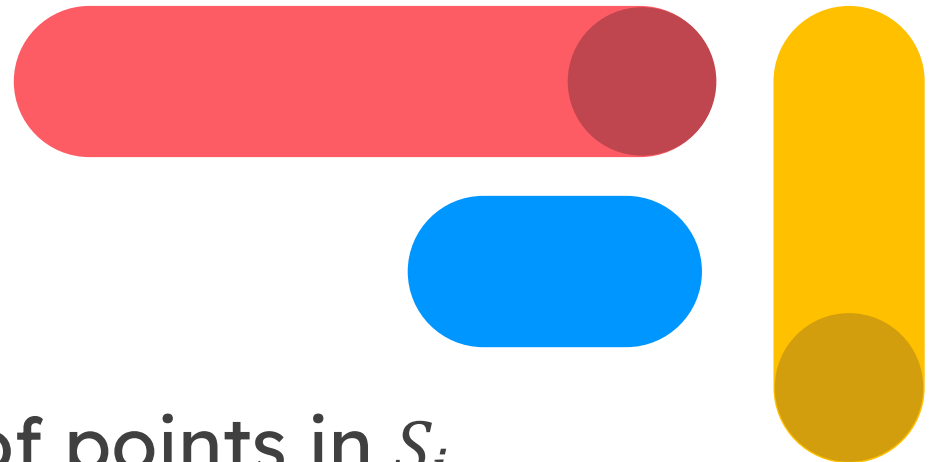


Given a set of observations $(x_1, x_2, \dots, x_n)$, where each observation is a d -dimensional real vector, k-means clustering aims to partition the n observations into k ($\leq n$) sets $S = \{S_1, S_2, \dots, S_k\}$ so as to minimize the within-cluster sum of squares (WCSS) (i.e. variance). Formally, the objective is to find:

$$\operatorname{argmin}_S \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2 = \operatorname{argmin}_S \sum_{i=1}^k |S_i| \operatorname{Var} S_i$$

where μ_i is the mean (also called centroid) of points in S_i

K-means Clustering



μ_i is the mean (also called centroid) of points in S_i

$$\mu_i = \frac{1}{|S_i|} \sum_{x \in S_i} x$$

$|S_i|$ is the size of S_i , and $\|\cdot\|$ is the usual L2 norm. This is equivalent to minimizing the pairwise squared deviations of points in the same cluster:

$$\operatorname{argmin}_S \sum_{i=1}^k \frac{1}{|S_i|} \sum_{x, y \in S_i} \|x - y\|^2$$

K-means Clustering

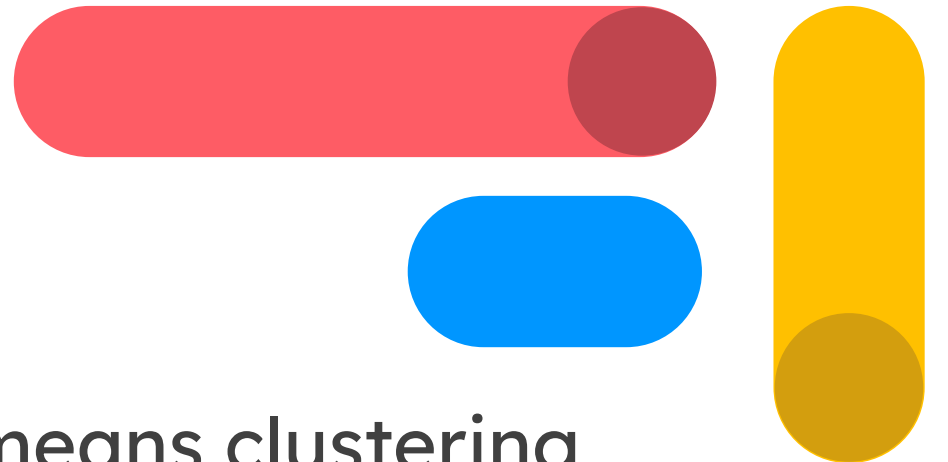
Since the total variance is constant, this is equivalent to maximizing the sum of squared deviations between points in different clusters (between-cluster sum of squares, BCSS).

Meanwhile, we minimize the differences within-cluster sum of squares (WCSS).

→ $\text{argmax}(\text{BCSS})$

→ $\text{argmin}(\text{WCSS})$

K-means Clustering



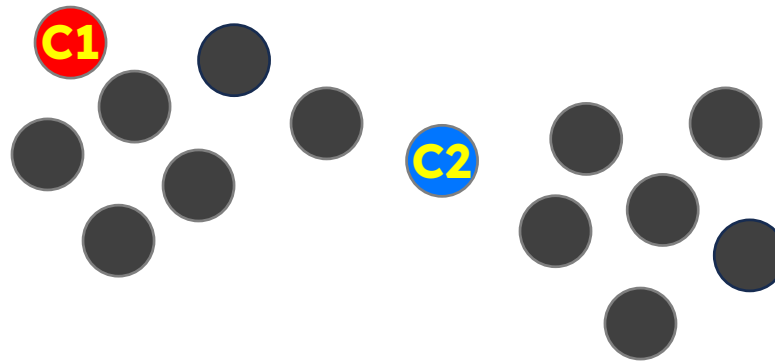
A step-by-step demonstration for k-means clustering

1 Define the number of clusters (k)

Given k value of 2

2 Select k random points from the dataset as centroids

The number of clusters is 2; therefore, we randomly select 2 data points as centroids

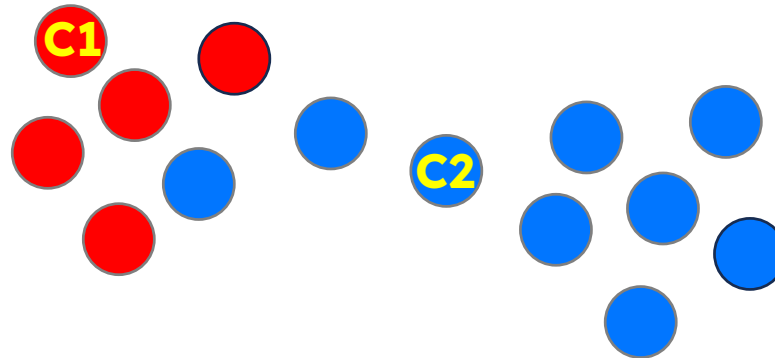


K-means Clustering

A step-by-step demonstration for k-means clustering

3 Assign all data points to the nearest centroid

For each data point has been assigned to a centroid by the closet distance

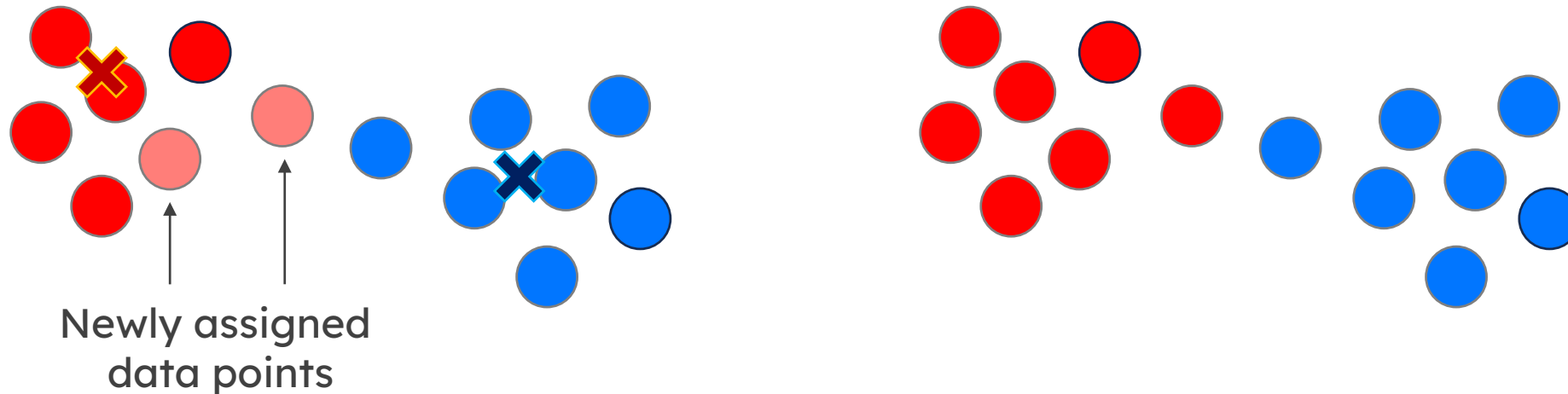


K-means Clustering

A step-by-step demonstration for k-means clustering

4 Update centroid by mean centers

Find the mean center of each group, defined by the previous step.

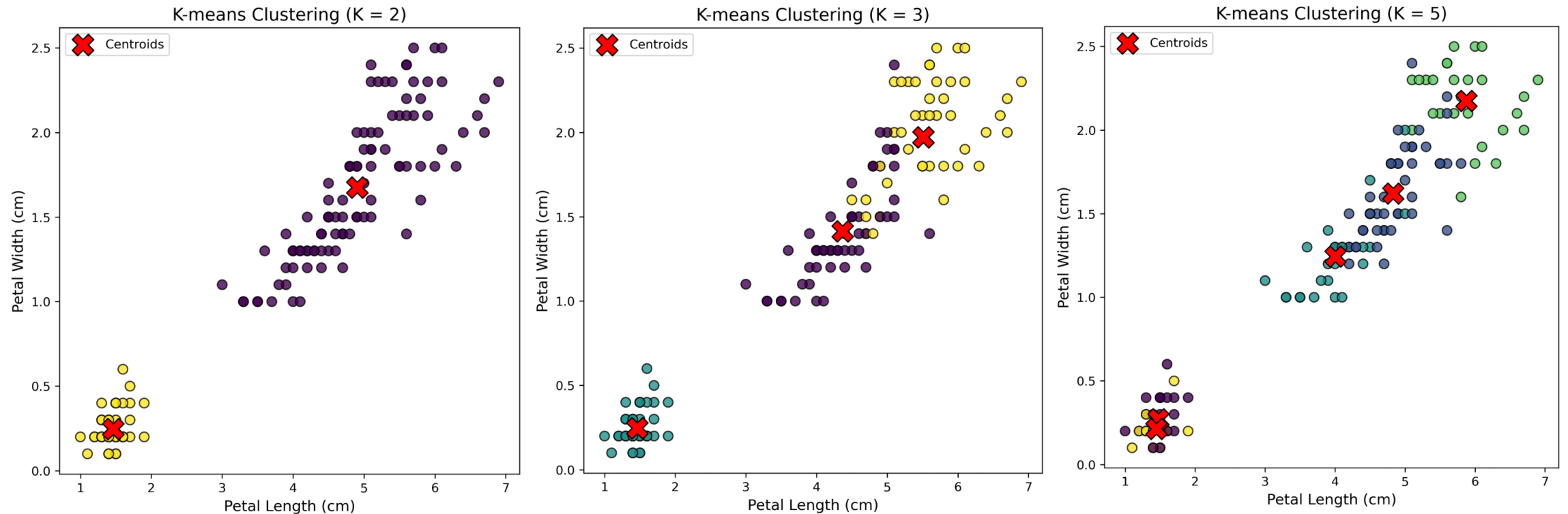


5 Repeat step 3 and step 4

- Centroids of newly formed clusters do not change
- Points remain in the same cluster
- Maximum number of iterations is reached

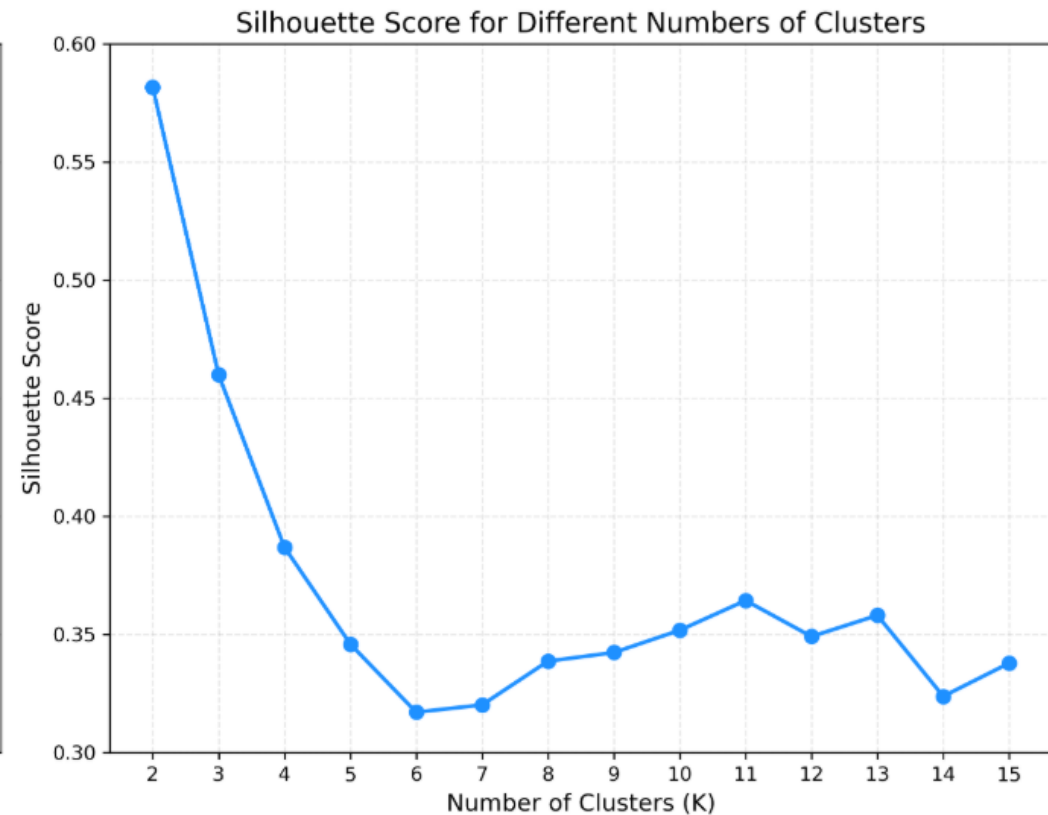
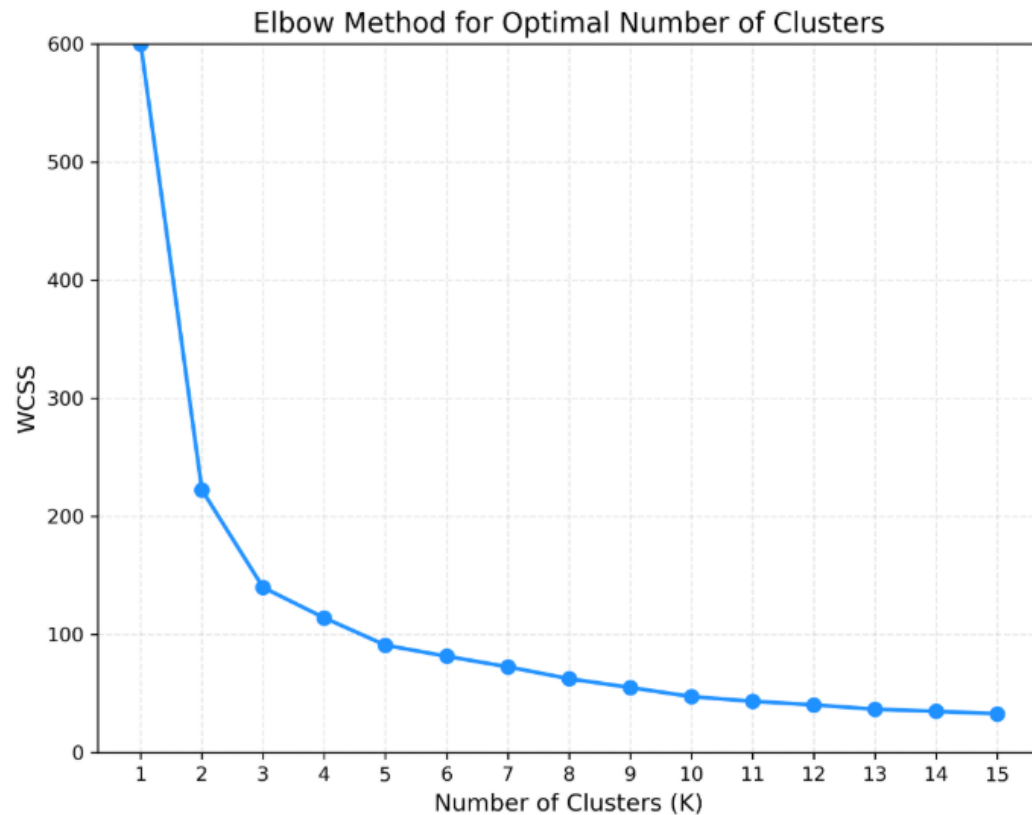
K-means Clustering

IRIS Demo Case for K-means Clustering

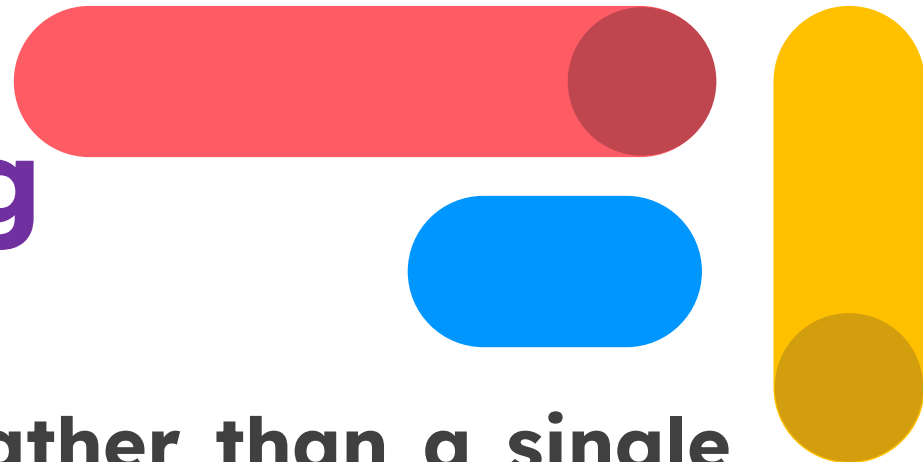


K-means Clustering

IRIS Demo Case for K-means Clustering



Hierarchical Clustering



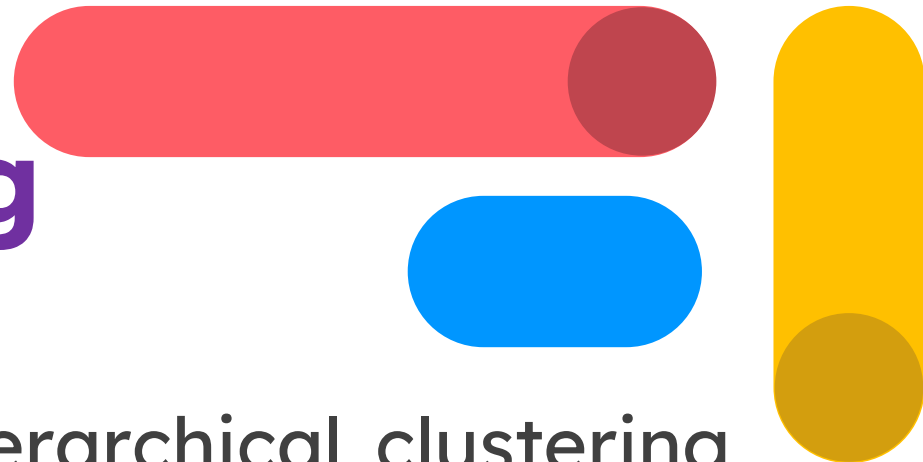
Building clusters as a hierarchy rather than a single partition

Hierarchical clustering progressively merges or divides observations according to their similarity, producing a nested structure of clusters.

- 1) No need to specify the number of clusters beforehand
- 2) Represents relationships through a dendrogram
- 3) Cluster formation depends on the linkage criterion

Common linkage methods: Single, Complete, Average, and Ward

Hierarchical Clustering



Typically, there are two types of hierarchical clustering based on grouping strategy:

Method	Agglomerative	Divisive
Initialization	A cluster for each observation	All observation are the same cluster
Process	Gradually grouping by similarity	Gradually dividing
Direction	Bottom-up	Top-down
Practical usage	Mainly used	Seldom used

Hierarchical Clustering

Agglomerative Clustering

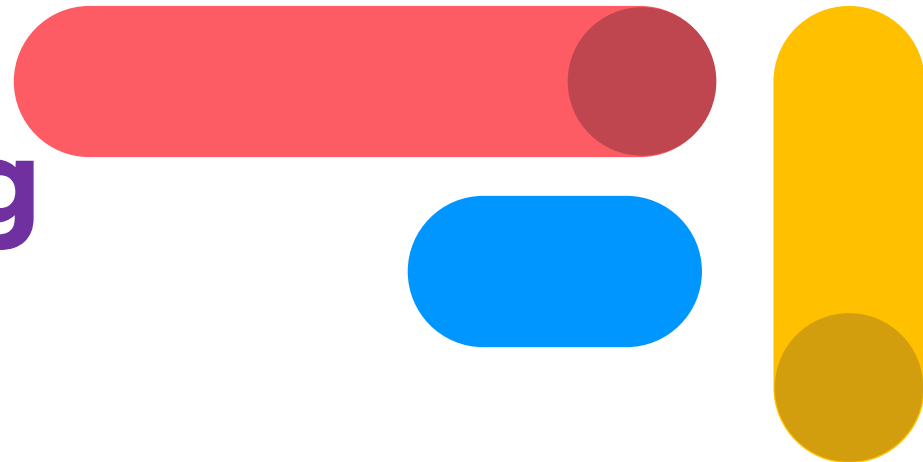
Given five observations {A, B, C, D, E}, then there are five clusters at the initial stage, i.e., {A}, {B}, {C}, {D}, and {E}.

a) Compute distance between two points

	A	B	C	D	E
A	0	1	5	6	8
B	1	0	4	5	7
C	5	4	0	2	6
D	6	5	2	0	5
E	8	7	6	5	0

$d(A, B) = 1$ is the nearest, so merge these two as {A,B}

Hierarchical Clustering



Agglomerative Clustering

b) Recompute distance between clusters

But, how to compute the distance between $\{A, B\}$ and $\{C\}$.

Given $Cluster_1 = \{A, B\}$, where $A = 1, B = 3$ and $Cluster_2 = \{C, D\}$, where $C = 6, D = 10$.

Cross-cluster distances:

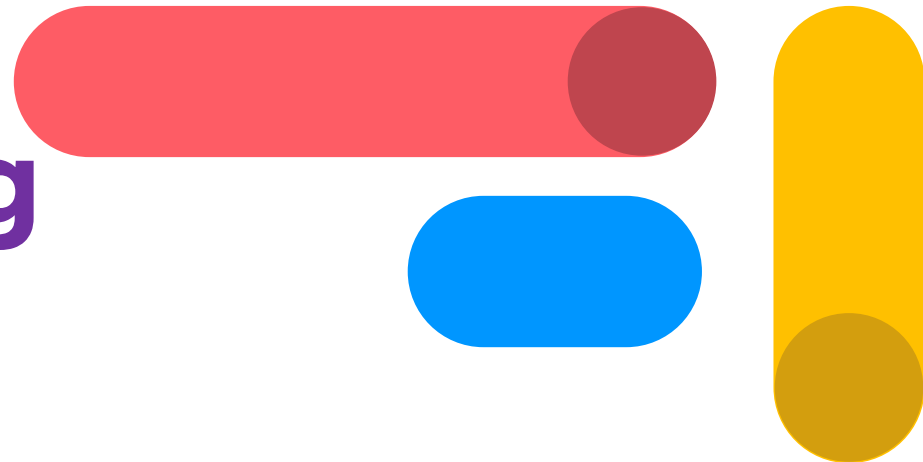
$$d\{A, C\} = 5; d\{A, D\} = 9; d\{B, C\} = 3; d\{B, D\} = 7$$

Single | $D = 3, \because \min(5, 9, 3, 7) = 3$

Complete | $D = 9, \because \max(5, 9, 3, 7) = 9$

Average | $D = 6, \because \text{mean}(5, 9, 3, 7) = 6$

Hierarchical Clustering



Distance Metrics

We need to define the distance between observations.

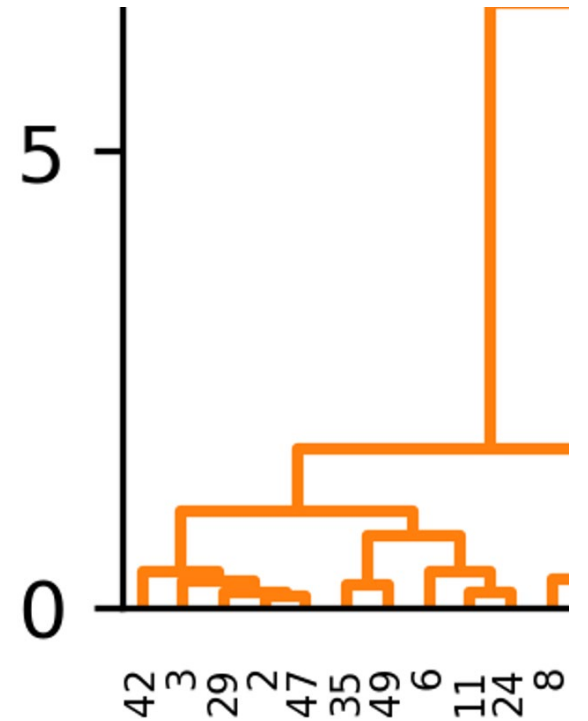
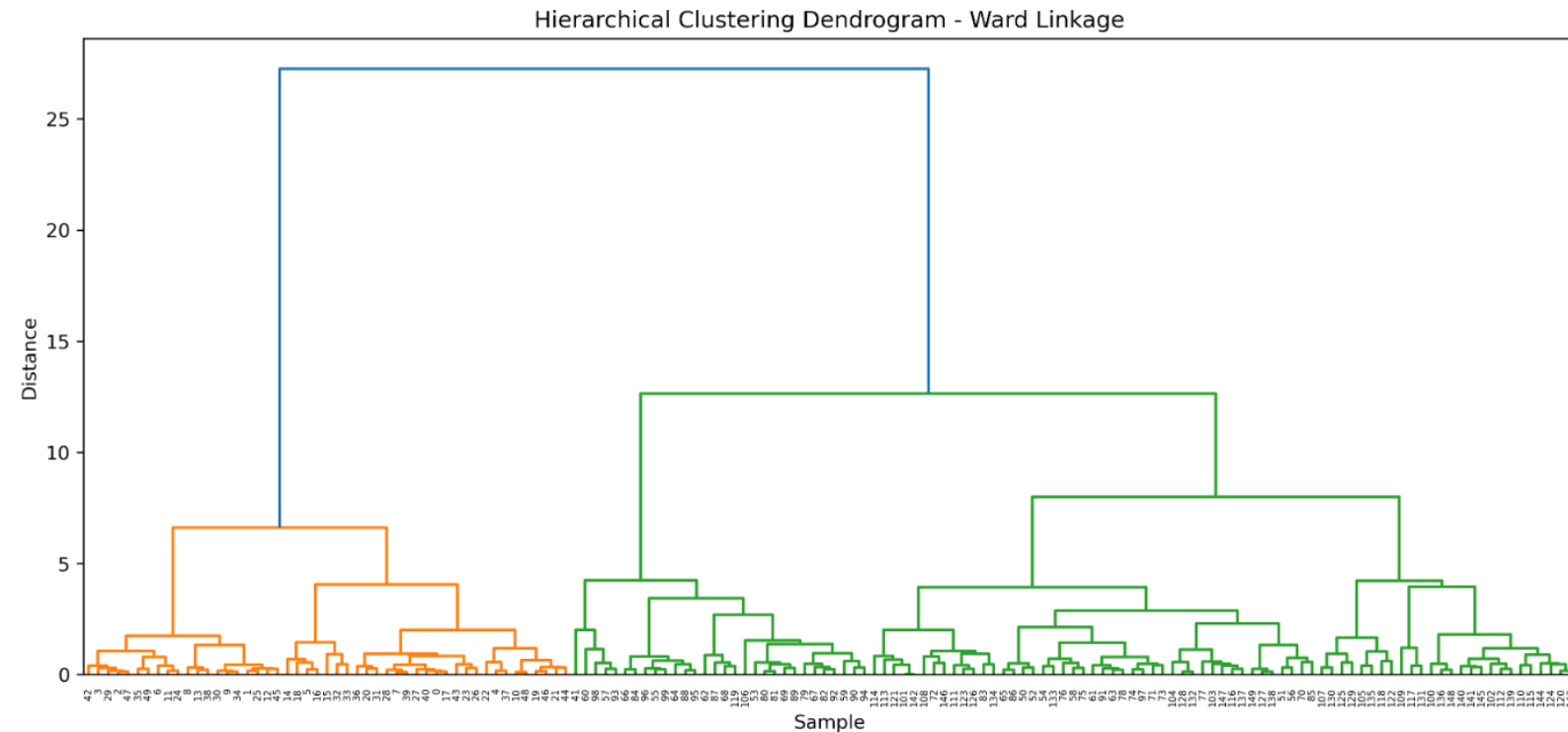
- 1) Manhattan distance (L1 norm)
- 2) Euclidean distance (L2 norm)
- 3) Correlation distance ($1 - \rho$)

Notice: Ward has to use Euclidean distance, since Ward is based on variance/ squared Euclidean distance.

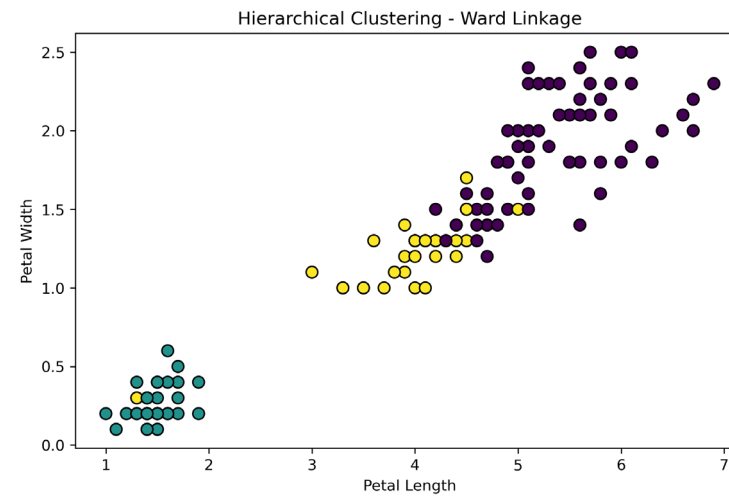
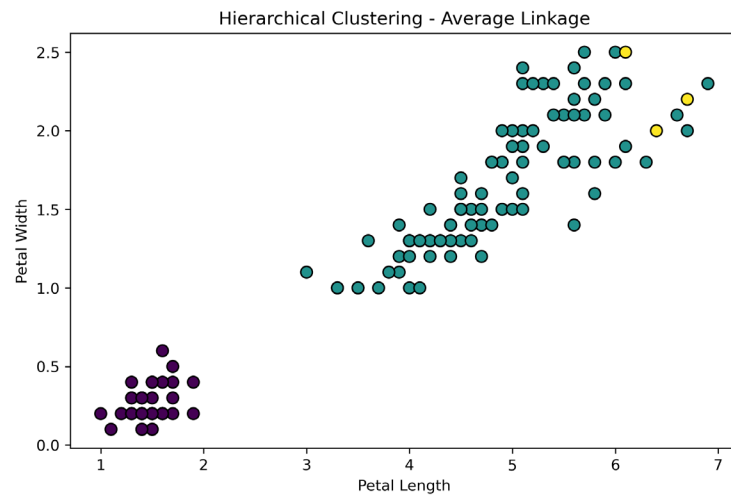
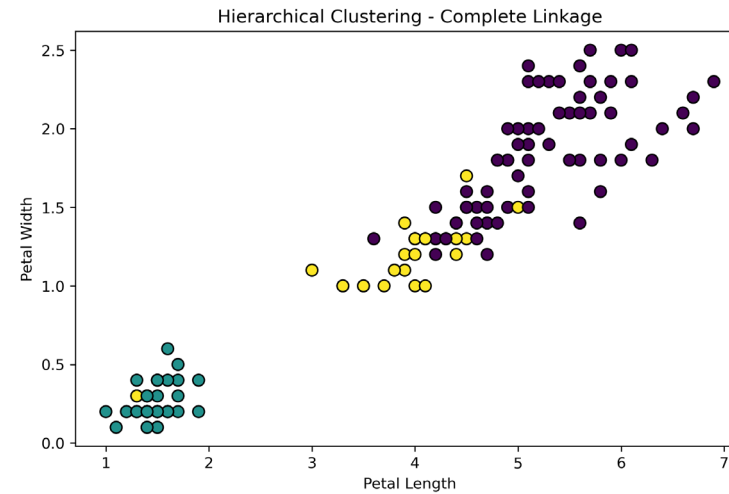
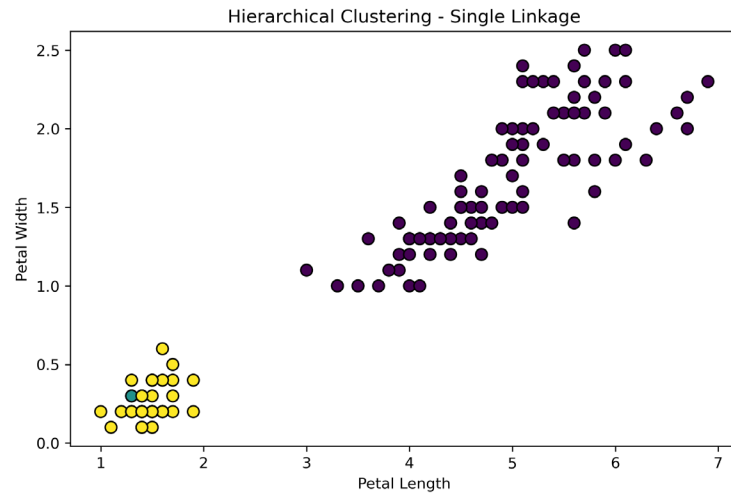
Hierarchical Clustering

Dendrogram

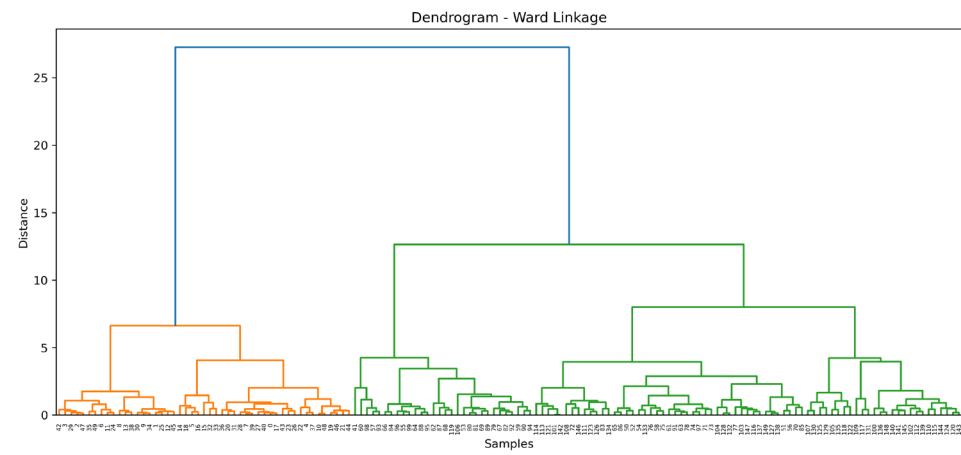
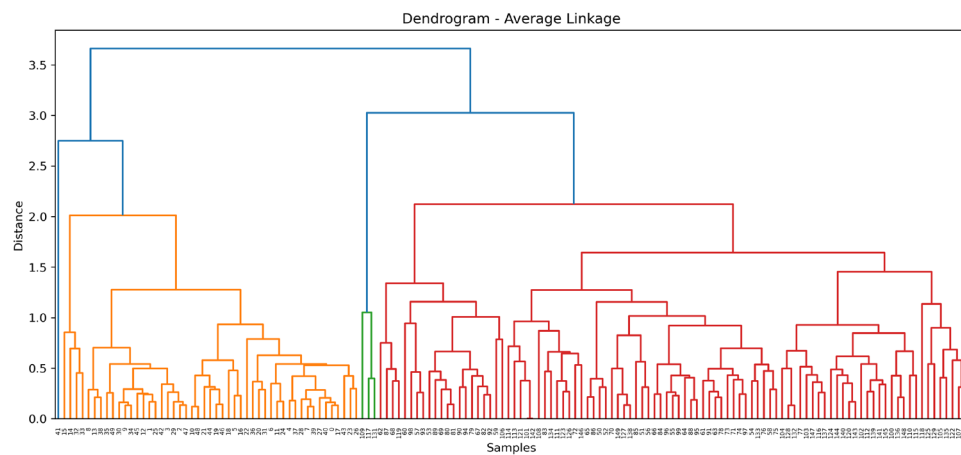
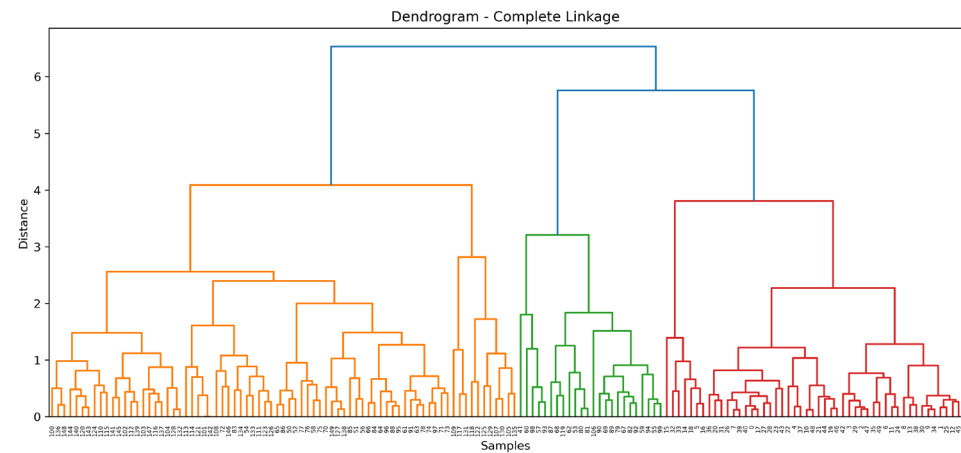
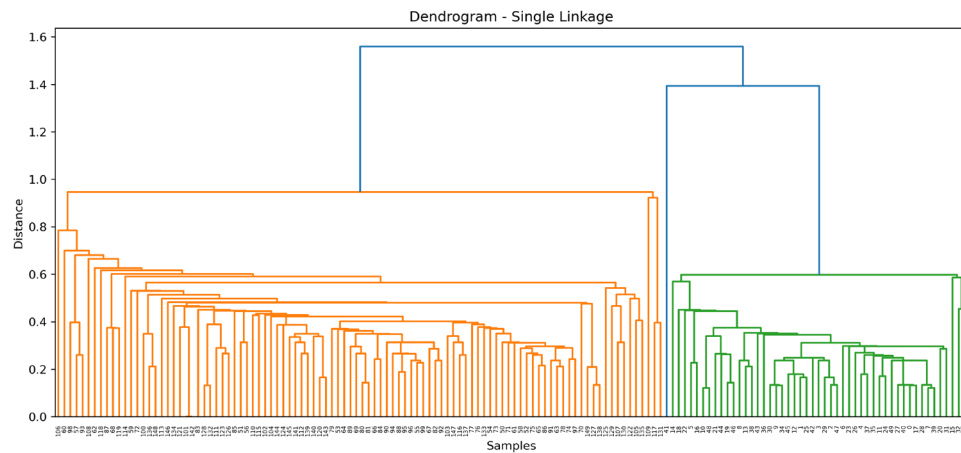
Distance at which cluster merge



Hierarchical Clustering



Hierarchical Clustering



DBScan

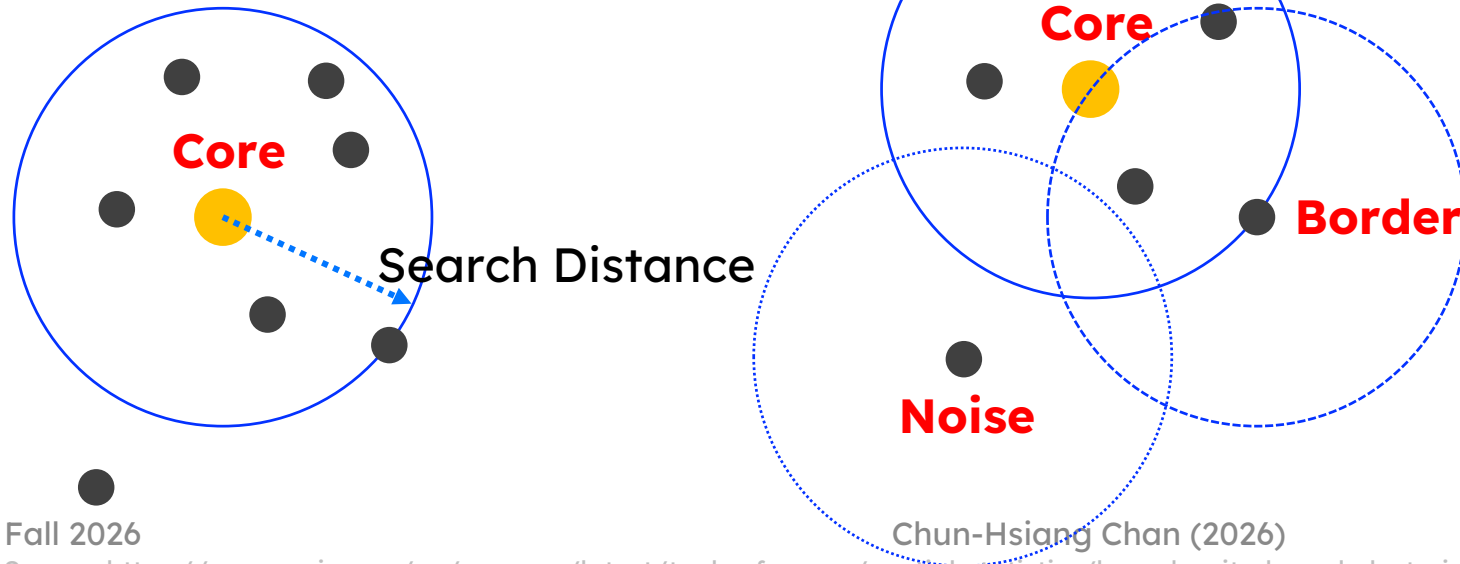
Finds clusters of point features within surrounding noise based on their spatial distribution. Time can also be incorporated to find space-time clusters.

This tool extracts clusters from the **Input Point Features** parameter value and identifies any surrounding noise.

Algorithm	Characteristics
DBSCAN	finds clusters of points that are in close proximity based on a specified search distance.
HDBSCAN	finds clusters of points similar to DBSCAN but uses varying distances, allowing for clusters with varying densities based on cluster probability.
OPTICS	orders the input points based on the smallest distance to the next point.

DBScan

The core-distance is related to the Search Distance parameter, which is used by both the Defined distance (DBSCAN) and Multi-scale (OPTICS) clustering methods.



1 Core Point

The number of points within the search distance of a **core point** is higher than the minimum points.

2 Border Point

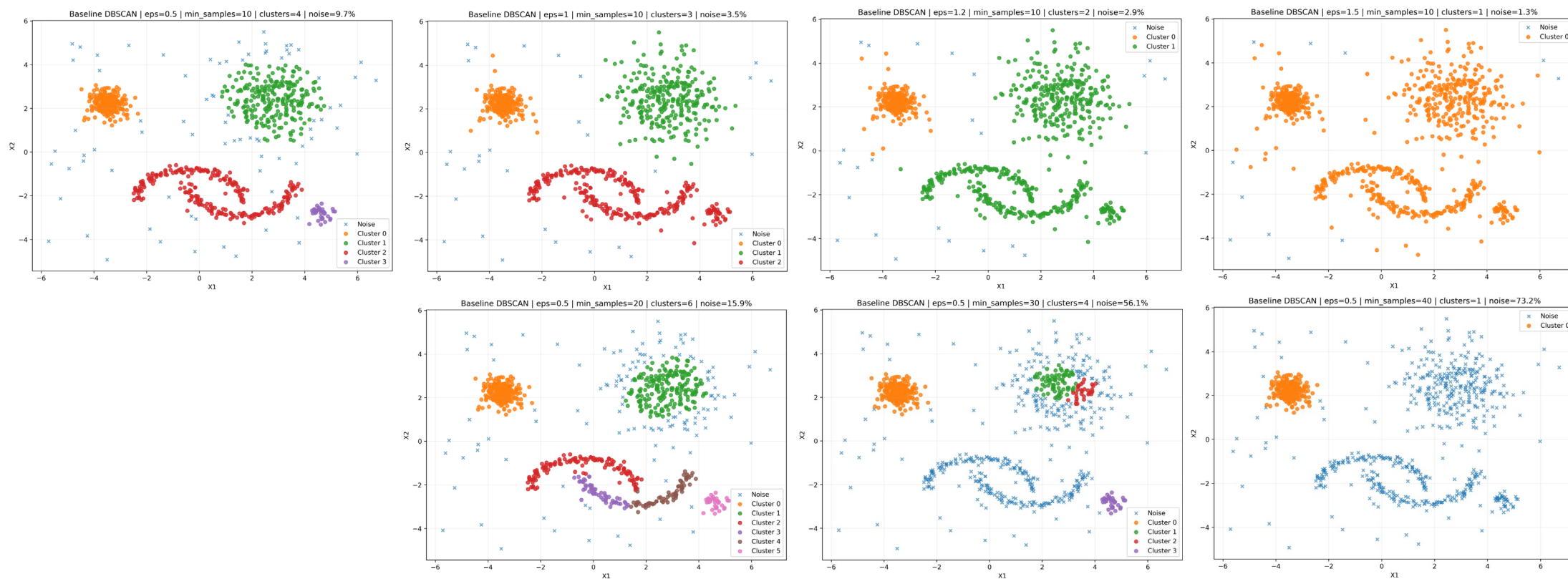
A **border point** is a point that is within the search distance of a core point but does not itself have the minimum number of features within the search distance.

3 Noise Point

If a point does not have the minimum number of features within the search distance and is not within the search distance of another core point

DBScan

Demo Case for DBScan



HDBScan

HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) can be understood as a hierarchical, advanced version of DBSCAN.

Its core value lies in eliminating the need to specify a fixed **eps** parameter upfront, unlike DBSCAN, it automatically identifies stable clusters across various density scales.



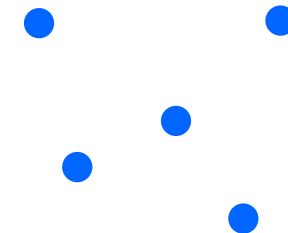
Too small eps

Clusters can
be found

Too large eps

Group all
together

All noise



Clusters can
be found

HDBScan

Instead of choosing one density threshold, examine clustering structures across multiple density levels.

To achieve this objective, here are four steps for HDBScan.

- 1) Estimate local density
- 2) Convert distance into mutual reachability distance
- 3) Build a hierarchy of clusters
- 4) Cluster stability

HDBScan

1) Estimate local density

K-nearest neighbor was adopted for measuring how close from one point to its neighbors.

We usually use core distance:

$$core_k(x) = d(x, k - th \text{ nearest neighbor})$$

If k -th nearest neighbor is very close, which represents the density of this location is high.

HDBScan

2) Convert distance into mutual reachability distance

HDBScan conducts the following formula to instead of Euclidean distance.

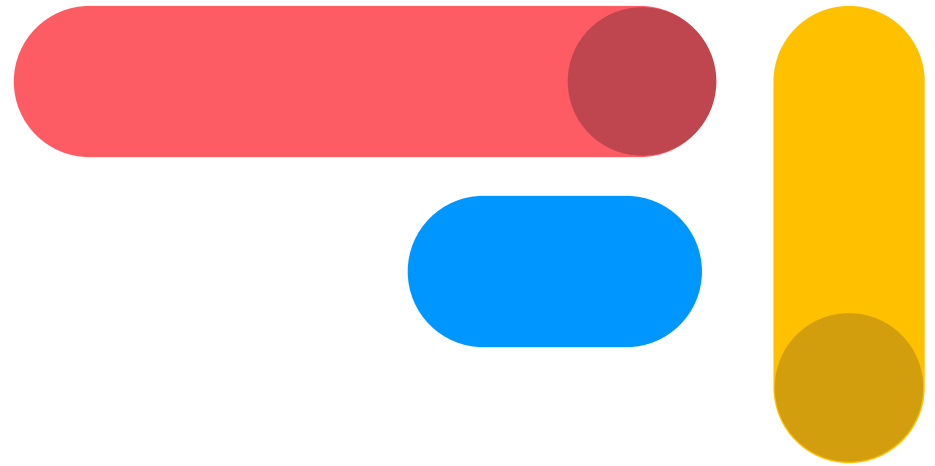
$$d_{mreach}(a, b) = \max\{core_k(a), core_k(b), d(a, b)\}$$

Maybe, there are a few points, which are close together, but they are just few points in a very sparse space.

Therefore, HDBScan uses the formula to avoid occasionally short distance between points, leading to formulate a incorrect cluster.

$$d(a, b) = 1; core(a) = 4; core(b) = 5; d_{mreach}(a, b) = \max(4, 5, 1) = 5$$

HDBScan

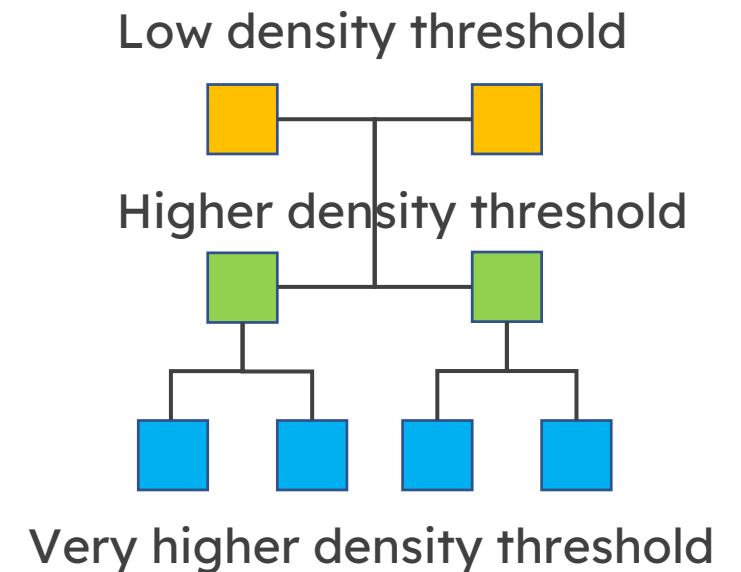


3) Build a hierarchy of clusters

HDBScan generates a minimum spanning tree to gradually change the density threshold.

As an increase in density, ...

- Large cluster will fall apart
 - Sparse connections will disappear first
 - Highly dense core keep remaining
- ➔ Cluster Hierarchy



HDBScan

4) Cluster stability

HDBScan should determine which level of clusters could be remained at the end of clustering analysis.

$$\lambda = \frac{1}{\text{distance}}$$

Shorter distance $\rightarrow$ higher density $\rightarrow$ larger λ (e.g., 2-10)

This cluster could be sustained in a wide density range, indicating it has higher stability.

$$\text{stability}(C) = \sum_{x \in C} (\lambda_{\text{leave}}(x) - \lambda(C))$$

HDBScan

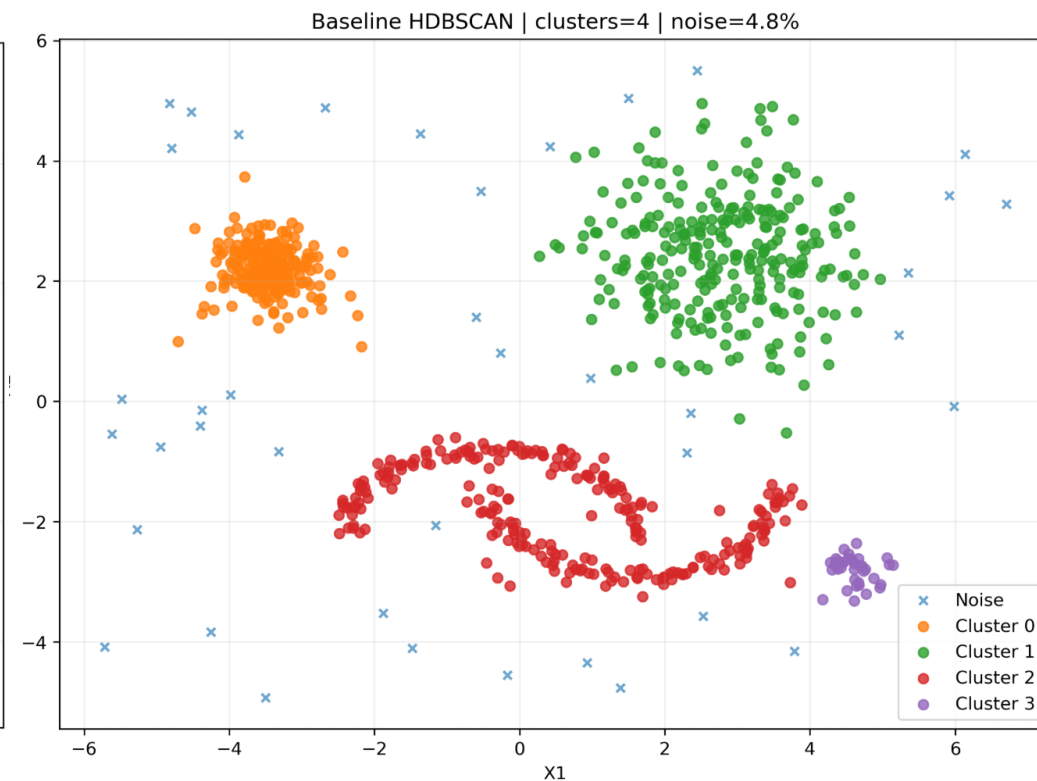
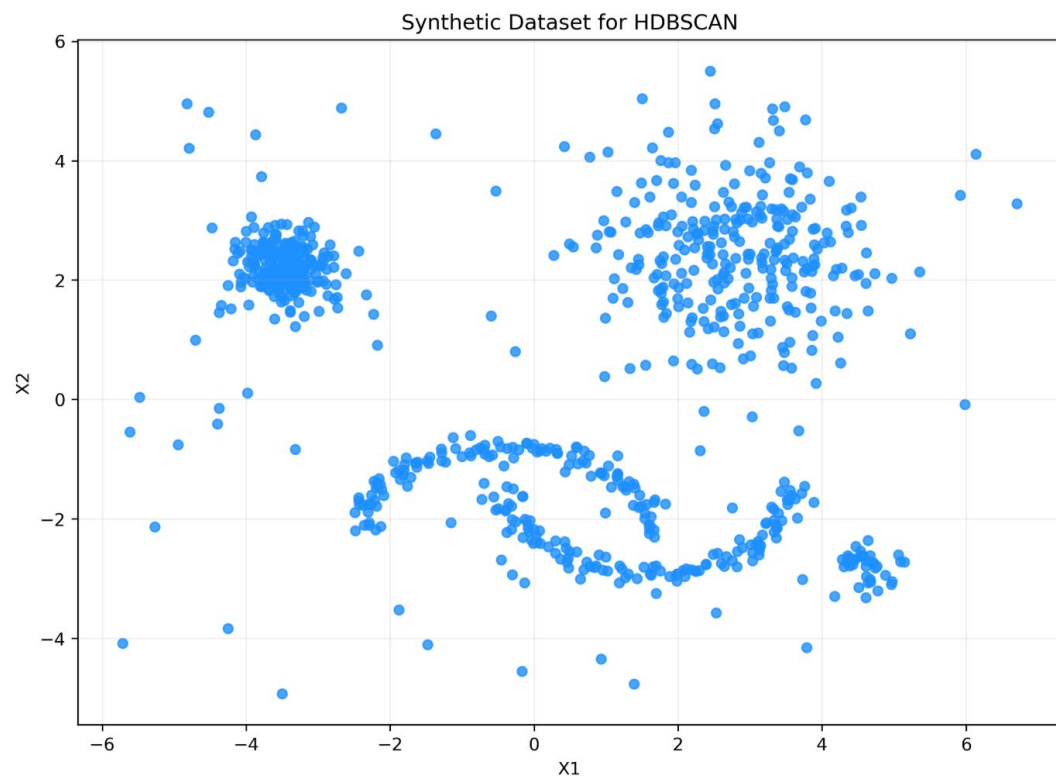
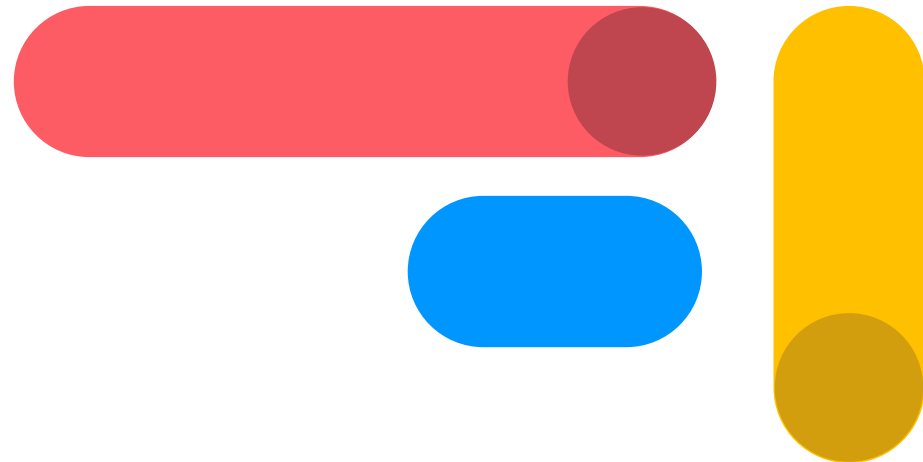
Hyperparameters

min_cluster_size: minimum sample size required for a single cluster

min_samples: controls how conservative the density estimation is.

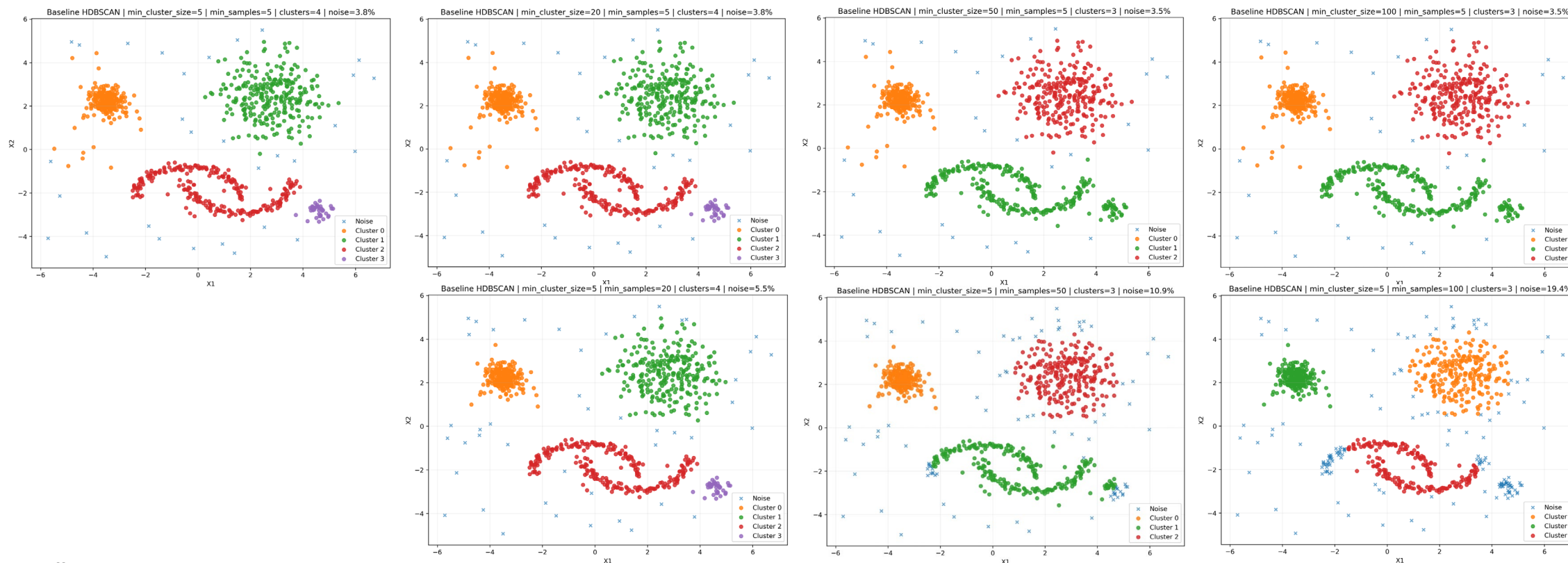
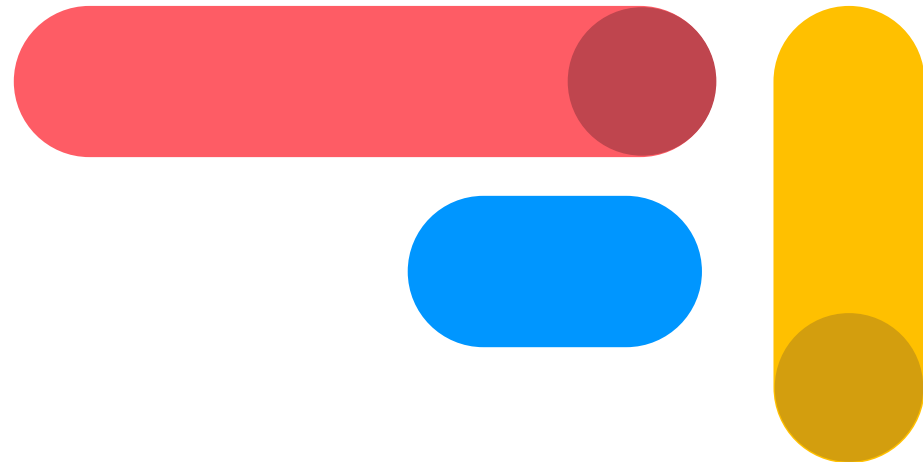
HDBScan

Demo Case for HDBScan

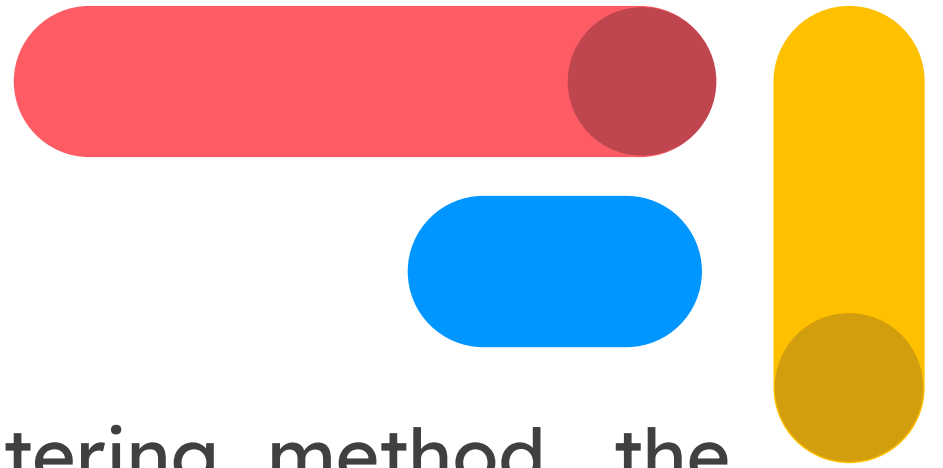


HDBScan

Demo Case for HDBScan



Optics



For the **Multi-scale (OPTICS)** clustering method, the search distance value is treated as the maximum distance that will be compared to the core distance.

OPTICS uses a concept of a minimum reachability distance, which is the distance from a point to its nearest neighbor that the search has not yet visited.

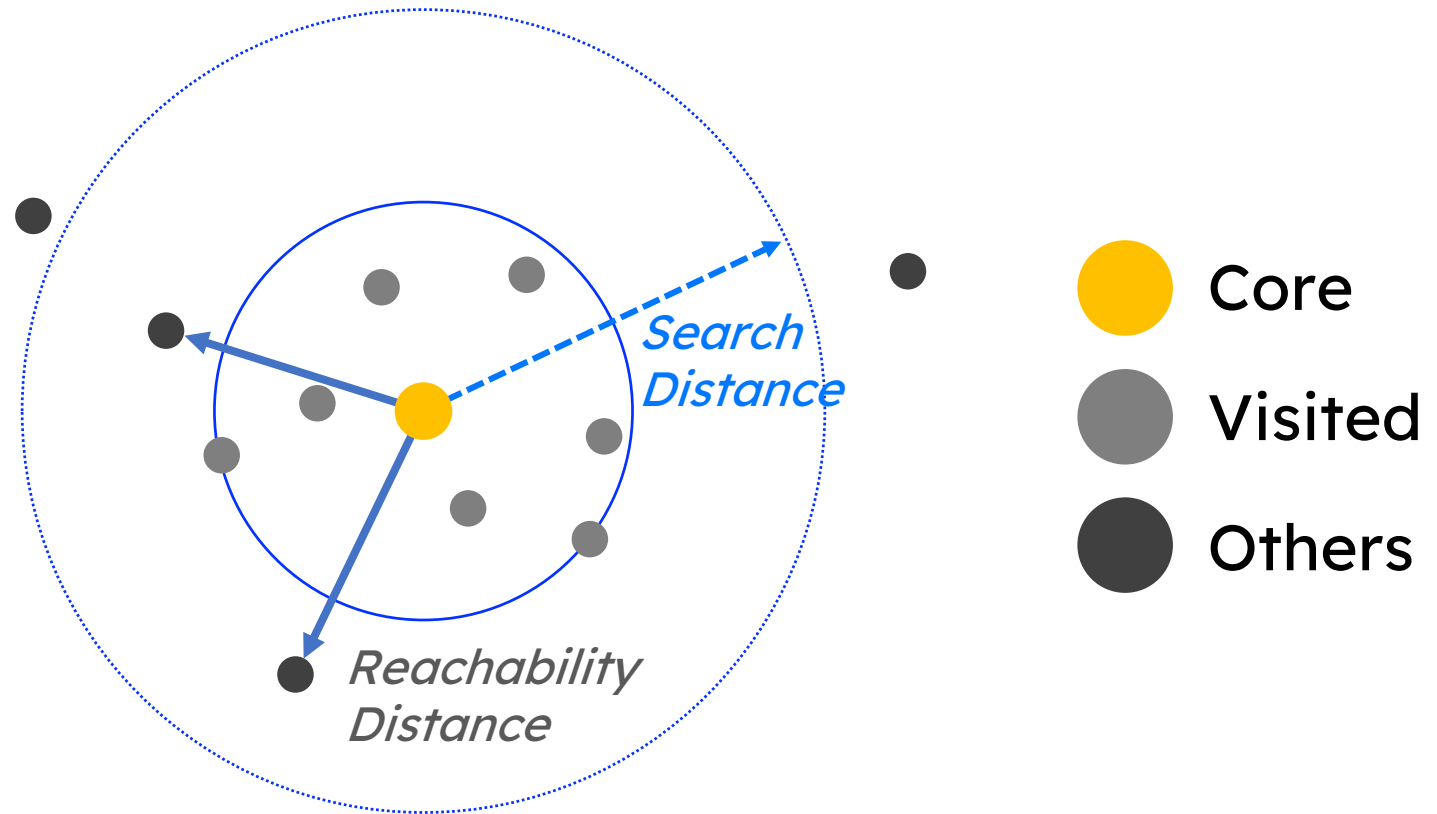
Optics

OPTICS will search all neighbor distances within the specified search distance, comparing each of them to the core distance.

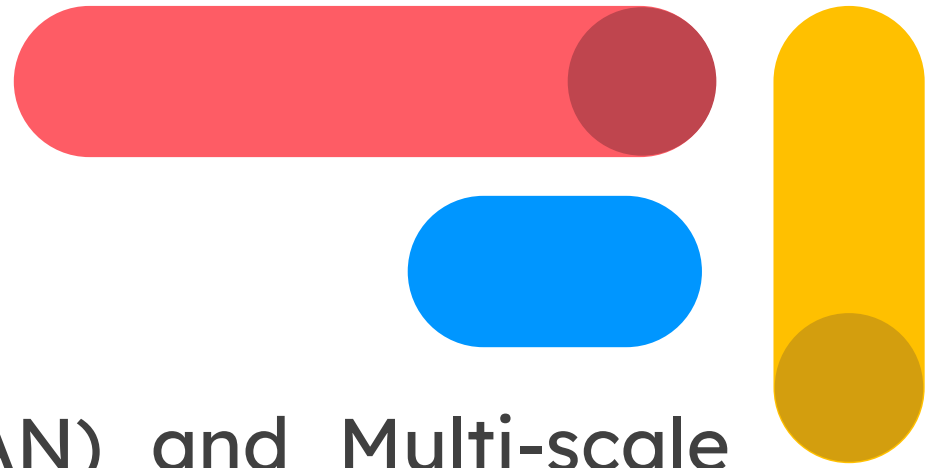
Situations	Core Distance Update	Core Distance
$\forall \text{NeighborDist} < \text{CoreDist}$	False	Original core distance
$\forall \text{NeighborDist} > \text{CoreDist}$	True	Replace by Min(Neighbor distance)

Optics

Revisit the situation again with a graph demonstration.



Optics



For both Defined distance (DBSCAN) and Multi-scale (OPTICS), **if no distance is specified, the default search distance is the highest core distance found in the dataset**, excluding those core distances in the top 1 percent (in other words, excluding the most extreme core-distances).

OPTICS is an ordered algorithm that starts with **the feature with the smallest ID and goes from that point to the next to create a plot**. The order of the points is fundamental to the results.

Optics

Optics follows the concept of DBScan, including core points, MinPts, neighborhood, and density connectivity.

1) Core Distance

Given a point p , if there should exist at least Minpts points, then ...

CoreDist(p) = distance from p to its MinPtsth nearest neighbor

If MinPts is 5 and the distance between p to 5th nearest points are [0.2, 0.3, 0.4, 0.7, 0.9, 1.3], then $CoreDist(p) = 0.9$.

→ How large radius can be covered at least MinPts points?

Short CoreDist, higher local density

Optics

2) Reachability Distance

If we need to discover point q from core point p , then

$$\text{ReachabilityDist}(q, p) = \max(\text{CoreDist}(p), d(p, q))$$

Given $\text{CoreDist}(p) = 0.8$ and $d(p, q) = 0.5$, then

$$\text{ReachabilityDist}(q, p) = \max(0.8, 0.5) = 0.8$$

Given $\text{CoreDist}(p) = 0.8$ and $d(p, q) = 1.2$, then

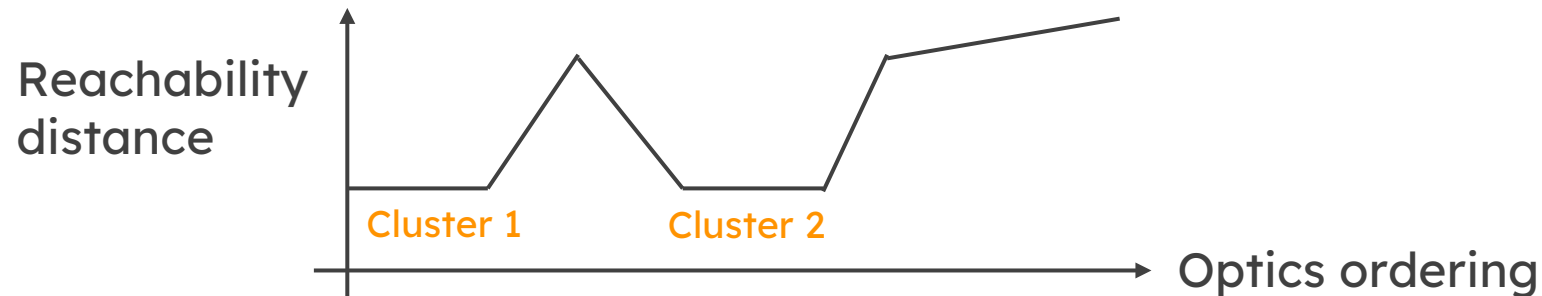
$$\text{ReachabilityDist}(q, p) = \max(0.8, 1.2) = 1.2$$

You have to extend 1.2 based on current density structure

Optics

How does Optics work?

- a) Ordering all data points (let density-connected points put together)
- b) Computing reachability distance for each point
- c) Illustrating reachability plot
- d) Determine the cluster members and numbers



Optics

Hyperparameters

min_samples: the minimum points for defining a cluster

max_eps: maximum search distance

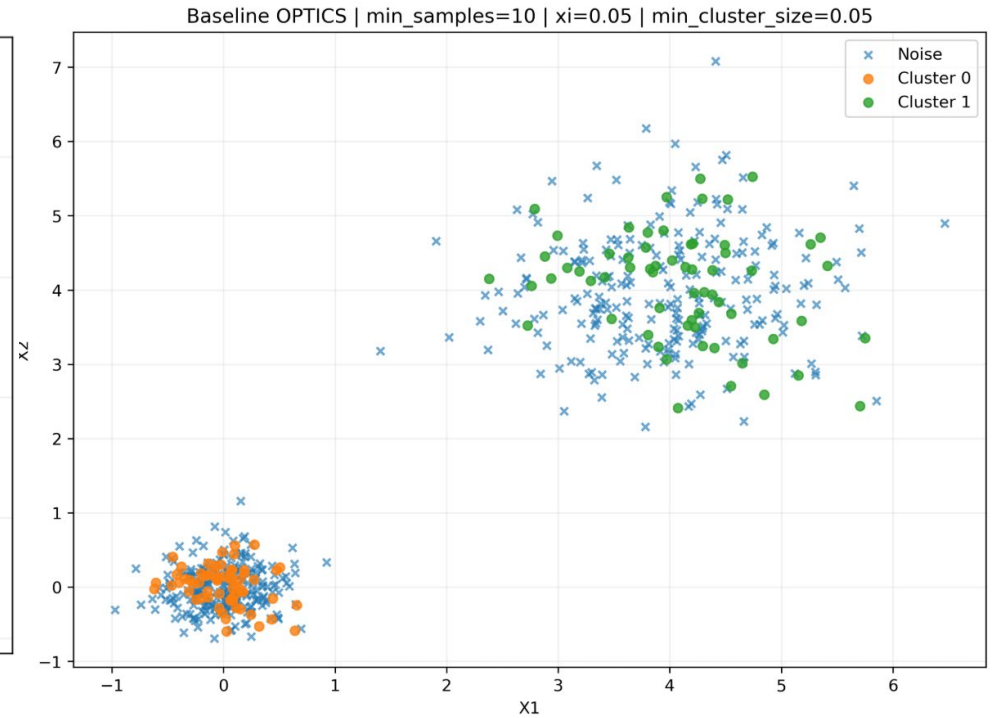
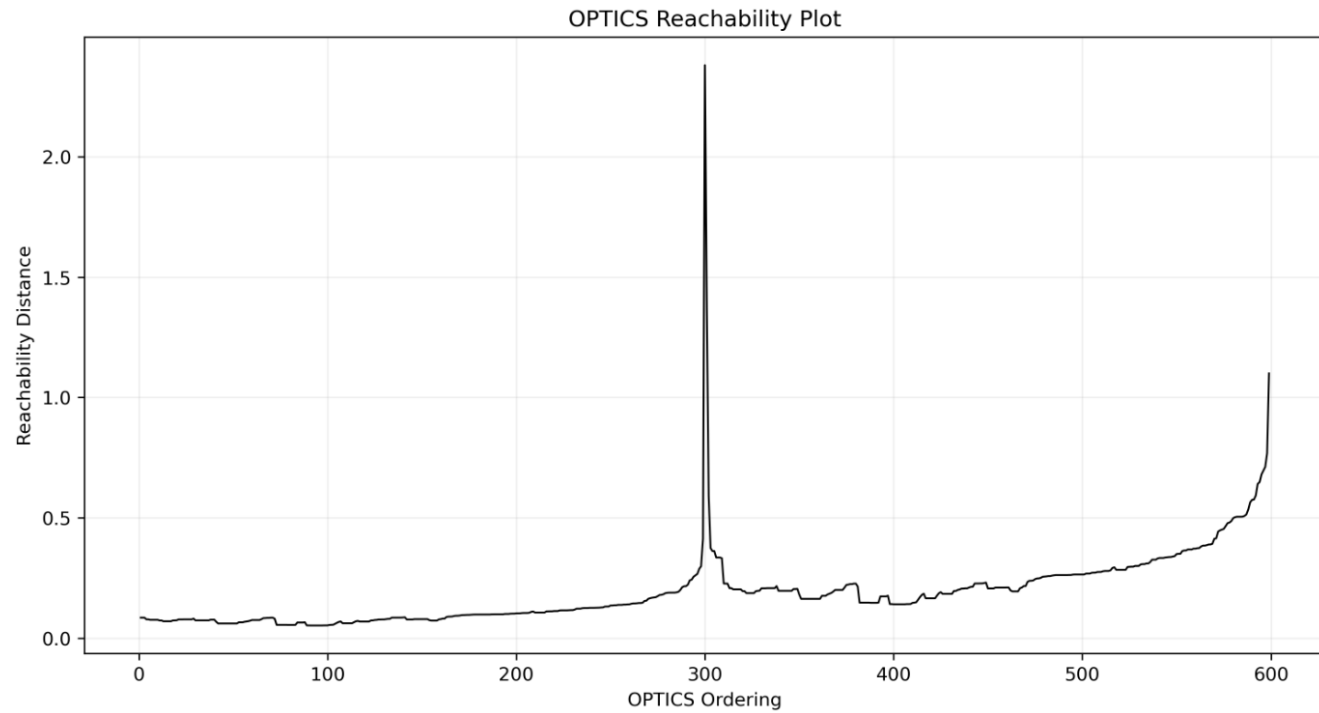
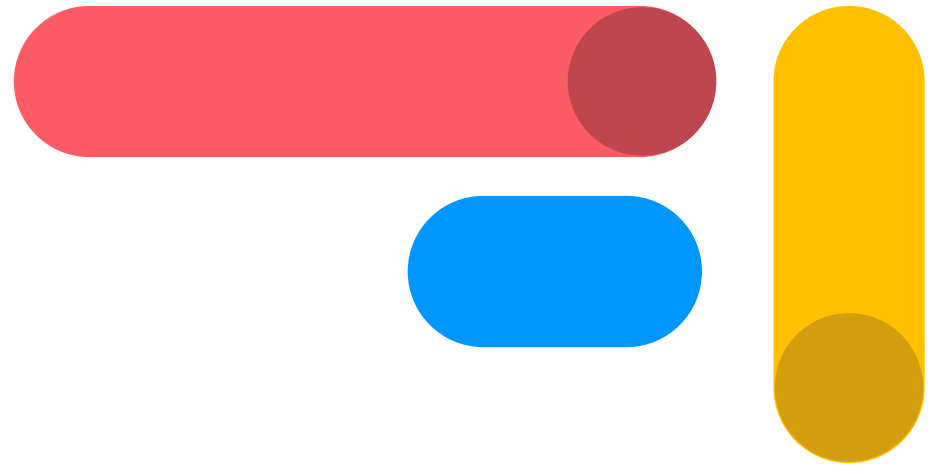
xi: the threshold for cluster boundary based on density changes; smaller xi, more clusters

min_cluster_size: the minimum points within a cluster.

If total samples $N=1000$, then sample size of each cluster will be at least $0.05 \times 1000 = 50$ points

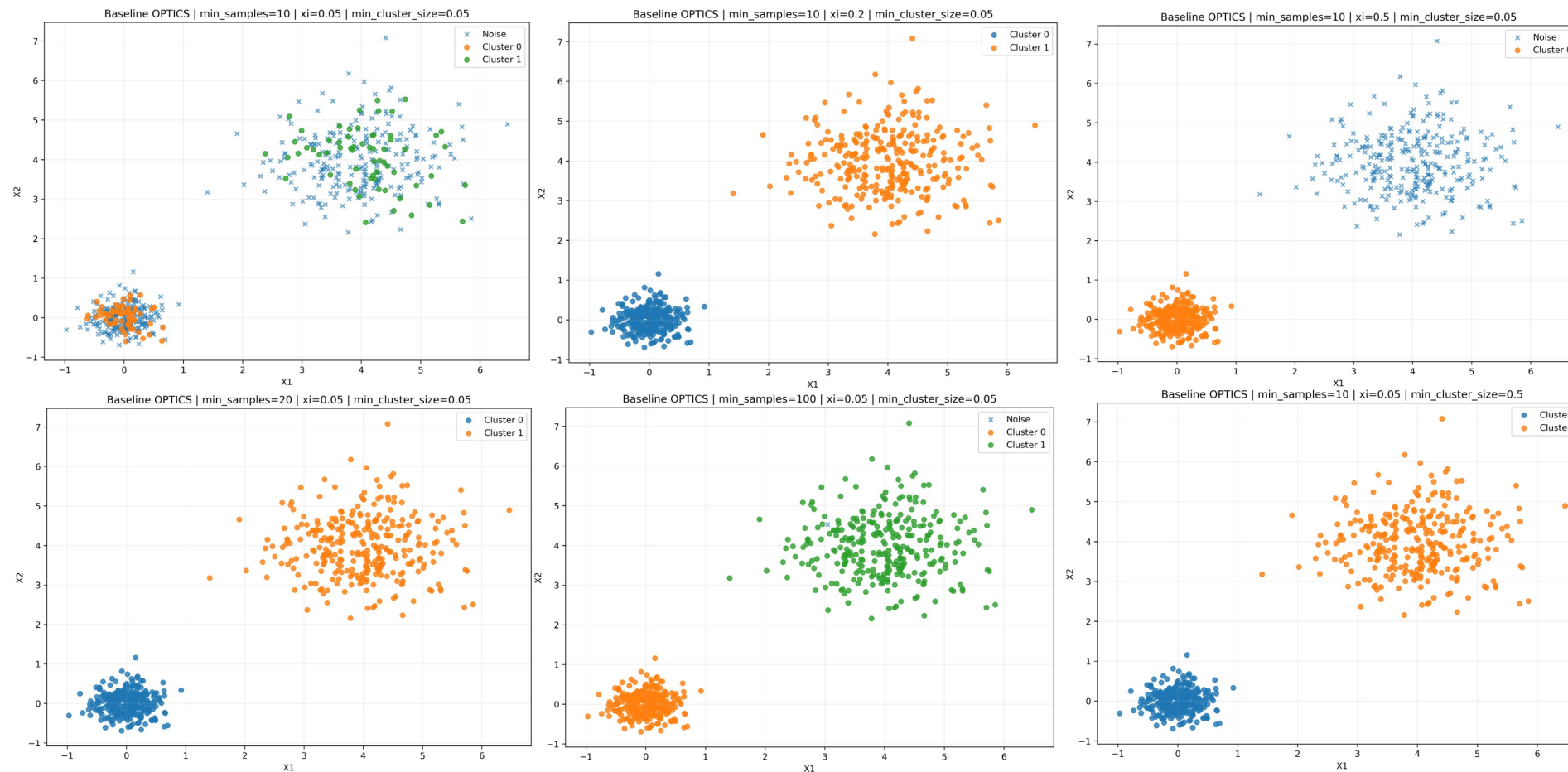
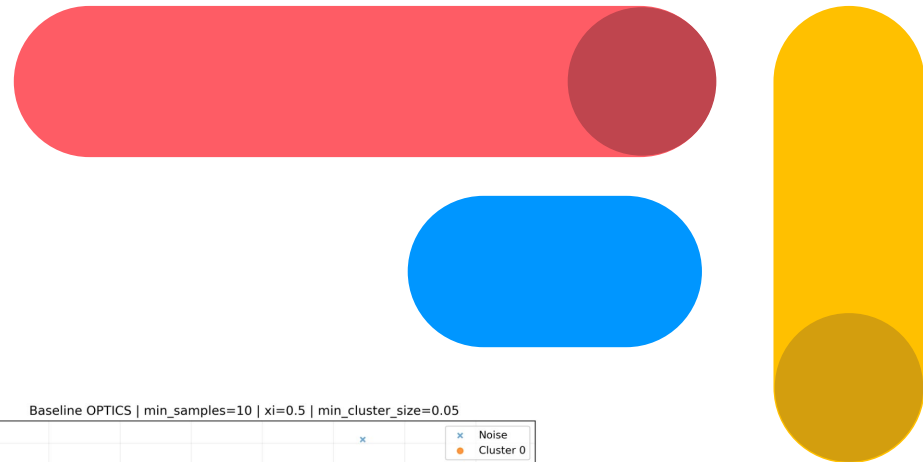
Optics

Demo Case for Optics

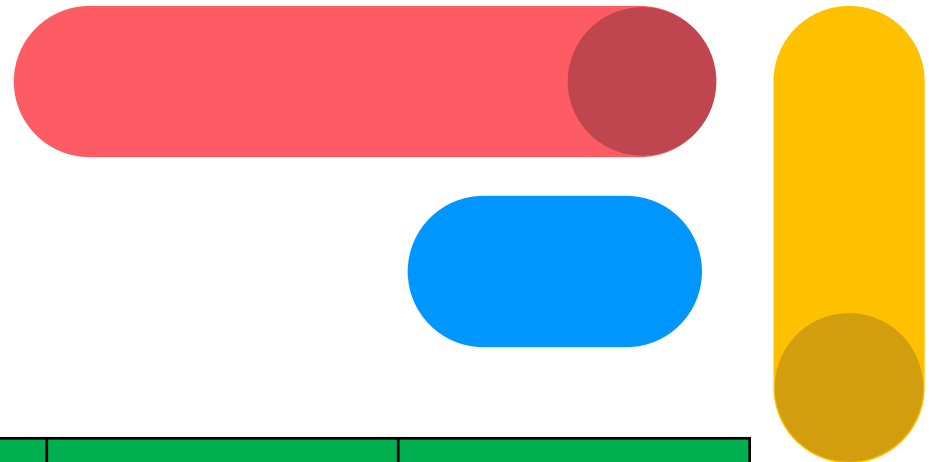


Optics

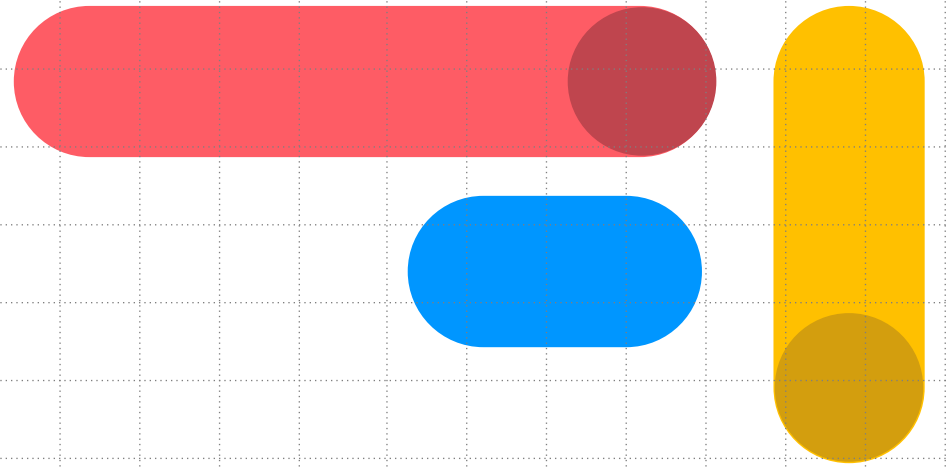
Demo Case for Optics



Summary



	K-means	Hierarchical	DBSCAN	HDBSCAN	OPTICS
Need K?	V	⚠	×	×	×
Arbitrary shape	×	⚠	V	V	V
Detect noise	×	×	V	V	V
Varying density	×	×	×	V	V
Hierarchy	×	V	×	V	⚠
Key output	Centroids	Dendrogram	Core / Border / Noise	Condensed tree	Reachability plot



The End

Thank you for your attention!

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